

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.		12
A) Describe point – to – point, straight line and contouring control systems of NC with the help of neat sketches.	CO1	6
B) Explain the desirable characteristics of CNC machine structure (bed). Describe the materials used for the same.	CO1	6
C) Discuss the recirculating ball screws used in CNC machines using schematic diagrams.	CO1	6
Q.2 Solve Any Two of the following.		12
A) What is indexable cutting tool insert? Explain its ISO designation with the help of suitable example.	CO2	6
B) Explain the use of G00, G01, G02 codes during CNC programming with the help of suitable examples.	CO2	6
C) Enlist the work holding devices used in CNC machines and explain any two in detail using schematic diagrams.	CO2	6
Q. 3 Solve Any Two of the following.		12
A) Describe Electric Discharge machining w.r.t. basic principle, mechanism of material removal, and applications.	CO3	6
B) Compare Laser Beam Machining with Electron Beam Machining.	CO3	6
C) Explain the Water Jet Machining process in detail using schematic diagram.	CO3	6

Q.4 Solve <i>Any Two</i> of the following.		12
A) Explain the basic principles involved in thermal spraying with the help of schematic diagram. What are the different types of thermal spraying methods?	CO4	6
B) Explain the basic principles involved in Physical Vapour Deposition process. Give the classification of this process and explain any one of those in detail.	CO4	6
C) Briefly describe the electroplating and electroless plating processes.	CO4	6
Q. 5 Solve <i>Any Two</i> of the following.		12
A) Explain the steps involved in rapid prototyping process.	CO5	6
B) Differentiate clearly between the subtractive and additive processes. Explain fused deposition modelling process in brief.	CO5	6
C) Explain the LIGA micro-fabrication method with the help of schematic diagrams.	CO6	6

***** End *****